Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/19/2019
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Effects of Low-Dose Radiation on Neurovascular Remodeling and Blood-Retina Barrier Function in Mice, Rats and Rabbits		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship
Start Date:	12/14/2017	**End Date:**	11/30/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2020 per NSSC information (Ed., 8/8/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sun, Shu-Wei Ph.D. (Loma Linda University)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	A recent report shows that more than 50% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. To date, the mechanisms behind these effects are not fully understood. These visual disturbances have been hypothesized to be related to increases in intraocular (IOP) or intracranial pressure (ICP). Modeling study showed that compromise in the integrity of the vascular blood-brain barrier (BBB) would serve to elevate ICP. Damage to microvascular endothelial cells induced by oxidative stress is one of the major factors thought to underlie the increased permeability of the blood-retinal barrier (BRB), which maintains a similar function in the eye for regulating intraocular pressure as the BBB in the cranium. Our knowledge about low-dose radiation risks with long-term spaceflight on retinal vasculature and their potential role in altering the BRB is very limited and inconsistent. The objectives of this application are to characterize the impact of radiation doses and qualities on retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and alteration of cell-cell interactions in the function of blood-retina-barrier (BRB). Archived ocular tissues samples at dose ranges from 0.1 Gy to 1 Gy with multiple particle types and energies across multiple animal species will be available for sharing from three NASA funded investigators. We will evaluate radiation–induced retinal vascular and tissue remodeling at multiple time points. Our study will elucidate the spectrum of radiation-induced structural and functional alteration in retinal BRB function and the mechanisms by which these changes are mediated. Together, our unique, integrative, and quantitative activities with advanced MRI imaging techniques will provide insight into the molecular mechanisms and pathways of low-dose radiation-induced mitochondrial oxidative damage on retinal cell-cell communication, BRB integrity, and vascular remodeling. The data from established model will provide unique criteria for risks of functional detriments associate with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data from established model will provide unique criteria for risks of functional detriments associated with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Task Progress:	Archived ocular tissues samples at dose ranges from 0.1 Gy to 1 Gy with multiple particle types and energies across multiple animal species were available for sharing from three NASA funded investigators. We have evaluated radiation-induced retinal vascular and tissue remodeling at multiple time points. Our study will elucidate the spectrum of radiation-induced structural and functional alteration in retinal BRB function and the mechanisms by which these changes are mediated. Together, our unique, integrative, and quantitative activities with advanced MRI imaging techniques provide insight into the molecular mechanisms and pathways of low-dose radiation-induced mitochondrial oxidative damage on retinal cell-cell communication, BRB integrity, and vascular remodeling. The data from established model provide unique criteria for risks of functional detriments associate with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures. Archived ocular tissues from irradiated mice, rats, and rabbits have been received from collaborators. Fixed ocular samples are sectioning for histochemical analysis for identifying biomarkers for oxidative stress, apoptosis, and blood-retina-barrier (BRB) function. Frozen retina samples were used for proteomics pathway analysis. Our data showed that space radiation conditions induce acute and long-term response in the retina. Mouse study demonstrates that ionizing radiation induces early apoptosis in mouse retina at a dose as low as 10 cGy. Exposure to low-dose space radiation induces oxidative stress associated apoptosis and changes in rabbit retinal protein expression related to cellular structure and function in the retina and retinal endothelial cells. Our data provide evidence that these cellular responses may affect visual acuity and impact the potential risk of developing late retinal degeneration. The data were presented at the NASA Human Research Program Workshop, 2019, and will be presented at Radiation Research Annual meeting 2019. Mice data was also published at Radiation Research 2018 (PMID: 29741442; reported last year) and Radiation Research 2019 (PMID: 30430917). Results Summary 1. Quantitative assessment of rabbit ocular tissue demonstrated that both protons and 16O induced significant apoptosis in the retina vascular endothelial cells compared to controls ($p<0.05$). The levels of 4-hydroxynonenal (4-HNE), a marker for lipid peroxidation, were significantly increased in the retina following proton and 16O ions exposure. 2. Proteomic analysis showed that many key pathways responsible for cell death, cell cycle, mitochondrial function, and metabolic stress were significantly altered in irradiated rabbit retina compared to age matched controls. 3. Prolonged exposure to neutron radiation induced significant changes in the rat global ocular structure. 4. There were significant increases in the expression of AQP-4 and a decrease in the expression of the BRB-related tight junction protein, Zonula occludens-1 (ZO-1) in the irradiated rats compared to controls. Effects of low-dose 16O and prolonged exposure of neutron irradiation on rat retina: The purpose of the current study was to characterize the impact of low-dose oxygen and prolonged exposure of neutron radiation on rat retina and retinal blood-retinal barrier (BRB) function. 6 month old male Long Evans rats were exposed to whole-body oxygen ions (16O, 1 GeV/n, 1cGy, and 10cGy) or sham-irradiation and the rats were followed up to 6 or 7 months after irradiation. For neutron irradiation, 6 month old male Wistar rats were exposed to whole-body neutrons over 6 months for total dose of 18cGy. Eyes were collected for histology and MicroCT analysis. Exposure to 16O caused significant dose-dependent increase of apoptosis in the retina and retinal endothelial cells. Levels of 4-hydroxynonenal (4-HNE) protein, an oxidative specific marker for lipid peroxidation were also elevated in the retina after radiation compared to controls. Immunohistochemical analysis of the retina revealed that an increased expression of aquaporin-4 (AQP-4) in the irradiated rats compared to controls gave strong indication of disturbance of BRB integrity. Prolonged exposure to neutron radiation induced significant changes in global ocular structure. The thickness of retina and choroid layer were significantly reduced in the irradiated rat eyes compared to controls. Quantitative assessment of ocular tissue demonstrated that the irradiated groups had significant increase of apoptosis and oxidative damage in the retina and retinal

	vascular endothelial cells compared to control groups. There was also a significant increase in the expression of AQP-4 and a decrease in the expression of the BRB-related tight junction protein, Zonula occludens-1 (ZO-1) in the irradiated rats compared to controls. In summary, our study revealed that exposure to low-dose ionizing radiation induced cellular oxidative damage that may alter retina structure and BRB integrity. Significant changes in retinal endothelial cells occur at doses as low as 1 cGy. These cellular responses may affect visual acuity and impact the potential risk of developing late retinal degeneration.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Mao XW, Nelson G, Jones T, Campbell-Beacher M, Stanbouly S, Nishiyama NC, Rodriguez D, Ortloff L, Sun S, Sridharan V, Boerma M, Hauer-Jensen M. "Vascular effects of space radiation." 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. Abstracts. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019
Abstracts for Journals and Proceedings	Mao XW, Boerma M, Nelson G, Shiba D, Shirakawa M, Takahash S, Delp M. "Impact of space flight or simulated microgravity combined with space radiation exposure on retinal oxidative damage." Presented at 40th International Society for Gravitational Physiology (ISGP) Meeting, Nagoya, Japan, May 26-31, 2019. Abstracts. 40th International Society for Gravitational Physiology (ISGP) Meeting, Nagoya, Japan, May 26-31, 2019. , May-2019
Abstracts for Journals and Proceedings	Mao XM, Nishiyama N, Stanbouly S, Davis CM, Britten R. "Effects of low-dose 16O and prolonged exposure of neutron irradiation on rat retina." 65th Annual Meeting of the Radiation Research Society, San Diego, CA, November 3-6, 2019. Abstracts. 65th Annual Meeting of the Radiation Research Society, San Diego, CA, November 3-6, 2019. , Nov-2019
Articles in Peer-reviewed Journals	Mao XW, Boerma M, Rodriguez D, Campbell-Beachler M, Jones T, Stanbouly S, Sridharan V, Nishiyama NC, Wroe A, Nelson GA. "Combined effects of low-dose proton radiation and simulated microgravity on the mouse retina and the hematopoietic system." Radiat Res. 2019 Sep;192(3):241-50. Epub 2018 Nov 15. https://doi.org/10.1667/RR15219.1 ; PubMed PMID: 30430917 , Sep-2019